Computer Systems

13th November 2024

Examination Paper

Answer ALL questions.

Clearly cross out any surplus answers.

Time: 3 Hours

The maximum mark for this paper is 100.

Any reference material brought into the examination room must be handed to the invigilator before the start of the examination.

Answer ALL questions

Marks

Part 1: Computer Systems and Hardware

1a) Describe TWO (2) main types of computer systems. **4**

1b) Explain the fundamental function of hardware **and** software in a computer system. **2**

1c) Explain the role of the operating system **and** its role in interfacing with hardware and software. Provide ONE (1) example of how the operating system works in this relationship. **2**

2a) Describe the function of THREE (3) modern computer hardware components – the CPU, Memory and I/O devices. **3**

2b) Explain the function of the ALU **and** control unit in the CPU. **2**

3a) Identify FOUR (4) features of a RISC that allow it to operate simply and efficiently. **4**

3b) Briefly describe ONE (1) advantage of a multi-core **and** ONE (1) advantage of a quantum computing systems. **3**

State which of these systems might be better to use for potentially breaking traditional encryption algorithms.

Total 20 Marks

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Part 2: Operating Systems

4a) State the FOUR (4) main components of an operating system. **4**

4b) State the purpose of timesharing and explain how it works as part of a virtual machine. **4**

Describe ONE advantage and ONE disadvantage of timesharing.

5a) State **and** describe TWO (2) examples of operating system resource management. **4**

5b) Explain the interaction between an operating system and hardware. **2**

5c) Outline FOUR (4) distinct characteristics of a RTOS (Real Time Operating System)'s operation that you may find in an embedded system such as a motorbike's electronic control unit (ECU). **4**

5d) Explain the role of virtual memory in an operating system **and** discuss how it improves system performance. **2**

Total 20 Marks

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Part 3: Numbers and Logic

6a) Convert the following decimal number to binary: **45** **2**
Show your working.

6b) Perform the following binary addition: 1010 + 0111 **3**

6c) Explain what is meant by binary coded decimal (BCD) **and** explain how it differs from binary. **4**

7a) Complete the following truth table for the logic function: A AND (B OR C). **9**

A	***B***	***C***		***A AND (B OR C)***
0	*0*	*0*	*0*	
0	*0*	*1*		
0	*1*	*0*	*1*	
0	*1*	*1*		*0*
1	*0*	*0*		*0*
1	*0*	*1*	*1*	*1*
1	*1*	*0*		
1	*1*	*1*	*1*	*1*

7b) Convert the hexadecimal number B37 to its decimal value. Show your working. **2**

Total 20 Marks

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Part 4: Software Hierarchy and Computer Programs

8a) Define the term "*addressing mode*" **and** state TWO (2) examples of addressing modes. **4**

8b) Explain the fetch-decode-execute cycle of instruction processing. **4**

8c) Describe an interrupt **and** explain how an interrupt mechanism works in computer systems. **2**

9a) Compare the roles of compilers **and** interpreters in software development. **6**

9b) Micro-controllers are widely used in several different application areas. For example, the automotive industry. **4**

Provide FOUR (4) additional examples of application areas where micro-controllers are used.

Total 20 Marks

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Part 5: Multiple Choice Questions (20 Marks)				
10a)	Which of the following best describes the function of an operating system.			**1**
	i.	To perform calculations and execute programs	ii. To manage computer hardware and software resources	
	iii.	To provide network connectivity	iv. To compile high-level programming languages	
10b)	What is the primary purpose of memory in a computer system?			**1**
	i.	To store the operating system only	ii. To execute arithmetic operations	
	iii.	To temporarily hold data and instructions for quick access	iv. To provide permanent data storage	
10c)	Which component of the computer system is primarily responsible for temporarily holding data and instructions for quick access during processing?			**1**
	i.	Control Unit	ii. Memory Unit	
	iii.	ALU	iv. Cache	
10d)	In data communications, which of the following is **NOT** considered a goal?			**1**
	i.	Reliability	ii. Security	
	iii.	Transmission efficiency	iv. Cost-effectiveness	
10e)	What is the binary representation of the hexadecimal value FC. Select ONE (1) option.			**1**
	i.	11111100	ii. 10100011	
	iii.	00110101	iv. 11001111	
11a)	Which network topology has all nodes connected to a central hub?			**1**
	i.	Bus Topology	ii. Ring Topology	
	iii.	Mesh Topology	iv. Star Topology	
11b)	Which of the following is a characteristic of quantum computing?			**1**
	i.	Uses classical bits for processing	ii. Utilises qubits to represent and process information	

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	iii. Operates on a binary logic framework	iv. Relies solely on traditional hardware components	
11c)	Which of the following types of software manages the hardware of a computer?		**1**
	i. Application software	ii. System software	
	iii. Middleware	iv. Firmware	
11d)	What is the primary purpose of the Hypertext Transfer Protocol?		**1**
	i. To send email messages between servers	ii. To establish a secure connection between two devices	
	iii. To facilitate the transfer of web pages and resources over the Internet	iv. To manage network traffic and routing	
11e)	Which device is used to connect multiple computers within a local area network (LAN) and manage data traffic between them?		**1**
	i. Modem	ii. Router	
	iii. Switch	iv. Hub	

Please turn over for the remaining questions

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12a)	Which of the following best describes the main purpose of the Control Unit (CU) in a CPU?				**1**
	i)	To perform arithmetic and logical operations	ii)	To control the flow of data and instructions within the CPU	
	iii)	To store frequently used data and instructions temporarily	iv)	To manage the input and output operations of the system	
12b)	Which system software is responsible for translating high-level programming code into machine language?				**1**
	i)	Database	ii)	Operating System	
	iii)	Compiler	iv)	Interpreter	
12c)	In which numbering system is the digit 'A' used to represent a value?				**1**
	i)	Binary	ii)	ASCII	
	iii)	Octal	iv)	Hexadecimal	
12d)	What is the main role of an operating system in a computer?				**1**
	i)	To manage hardware and software resources	ii)	To execute and compile programs	
	iii)	To connect computers in a network	iv)	To encrypt data for security	
12e)	Which of the following represents a type of computer network that spans a city or metropolitan area?				**1**
	i)	LAN	ii)	WAN	
	iii)	MAN	iv)	PAN	
13a)	What function does the Arithmetic Logic Unit (ALU) perform within a CPU?				**1**
	i)	Controls memory access	ii)	Executes arithmetic and logical operations	
	iii)	Manages data storage	iv)	Oversees the control flow of programs	
13b)	Which of the following is an example of embedded systems software?				**1**
	i)	Desktop operating system	ii)	Web browser	
	iii)	Firmware in a microwave	iv)	Office productivity software	

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13c)	The primary function of memory in a computer system is to:				**1**
	i)	Process data	ii)	Communicate with external devices	
	iii)	Control data flow	iv)	Store data temporarily or permanently	
13d)	Which of the following is NOT a benefit of RISC architecture over CISC?				**1**
	i)	Simplified instruction set	ii)	Faster instruction execution	
	iii)	Larger number of addressing modes	iv)	Reduced power consumption	
13e)	Which communication protocol is commonly used to ensure reliable data transmission over a network?				**1**
	i)	TCP	ii)	FTP	
	iii)	HTML	iv)	SMTP	
					Total 20 Marks

End of paper

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